



Reactive transport of different dissolved organic nitrogen components in an unconfined aquifer

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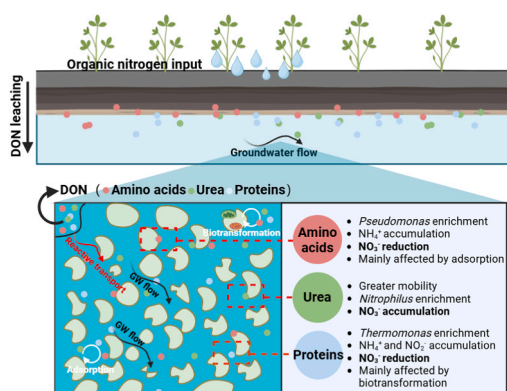
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HIGHLIGHTS

- Reactive transport mechanisms of DON in aquifers are clarified.
- Urea exhibits greater mobility, indicating a higher nitrogen contamination risk.
- Amino acids and proteins can cause a reduction of $\text{NO}_3\text{-N}$ concentration in aquifers.
- DON components were affected by biotransformation and adsorption to varying extents.

GRAPHICAL ABSTRACT



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ABSTRACT

Dissolved organic nitrogen (DON) is often an overlooked form of nitrogen that can leach from the soil into aquifers. The reactive transport and dispersion of DON in aquifers can contribute to regional nitrogen contamination. The current body of research has primarily focused on the vertical leaching process of DON through the vadose zone. However, these studies have largely ignored the broader reactive transport of DON within aquifers under the influence of groundwater flow. In this study, we investigate the reactive transport of DON under groundwater flow conditions. Utilizing molecular biological technologies, we aim to reveal DON's intrinsic role in the nitrogen cycle within aquifers. Our findings reveal that urea exhibits greater mobility compared to amino acids and proteins. The transport of amino acids and proteins reduces the $\text{NO}_3\text{-N}$ concentrations (44.6 % and 89.6 %) compared to the blank control, while urea leads to the accumulation of $\text{NO}_3\text{-N}$ in groundwater (10.1 %). Amino acid and protein columns show higher relative abundances of *Pseudomonas* (10.1 % and 7.3 %) and

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Thermomonas (3.9 % and 5.1 %) with denitrification functions, facilitating denitrification in groundwater. Conversely, the presence of urea increases the relative abundances of *Nitrosomonadaceae* and *Nitrophilus* (0.33 % and 0.67 %), posing a potential NO₃-N contamination risk. Biotransformation has the greatest effect on protein transport (19.6 %), while adsorption mainly influences amino acid transport (12.4 %). The study provides fundamental insights into the reactive transport of different DON components in aquifers, which holds important implications for regional groundwater environment protection.

1. Introduction

In recent decades, NO₃-N pollution in groundwater has emerged as an important global environmental challenge that impedes the sustainable utilization of water resources and regional economic growth [27]. Groundwater NO₃-N pollution is mainly caused by excessive agricultural fertilizer use, urban and industrial waste discharge, and exacerbated by extreme weather events that facilitate NO₃-N leaching into aquifers [35, 38,12]. Given the widespread presence of NO₃-N in groundwater, extensive research has been conducted on the transport and attenuation of NO₃-N in aquifers. For example, Carrey et al. [1] investigated the natural attenuation of NO₃-N during migration in groundwater and identified key factors controlling this process. Li et al. [19] explored how flow rate, nitrogen pollution load, and burial depth influence NO₃-N transport in groundwater.

Previous studies consider that the leaching of NO₃-N from soil substantially contributes to groundwater NO₃-N pollution [47,49], however, the presence and mobility of dissolved organic nitrogen (DON) cannot be overlooked. DON is also an important form of nitrogen in soil, consisting of proteins, peptides, amino acids, and other organic components [16]. The primary sources of DON include organic nitrogenous fertilizers, such as urea-based fertilizers and animal manure [51], as well as atmospheric nitrogen deposition and wastewater discharge [31]. In the soil, the transformation processes involving DON are highly complex. DON is not only a product of the breakdown of large organic molecules but also serves as a substrate for mineralization [17]. From the perspective of leaching, the loss of DON through leaching is another important pathway for nitrogen loss from the soil. Although previous studies have suggested that NO₃-N is the dominant form of nitrogen leaching due to its high mobility [33], recent findings indicate that DON leaching into aquifers can also be substantial [50]. For instance, in agricultural regions, DON leaching can account for up to 40 % of the total nitrogen leached [14]. Similarly, the amount of DON leaching from the forest area is usually high and increases with greater precipitation [26,32]. Therefore, it is essential to have a clear understanding of the fate of DON in soils in order to better elucidate its dynamics in groundwater and its impact on groundwater nitrogen contamination. Numerous site investigations and indoor tests have also demonstrated that the vadose zone (especially deep vadose zone) is an important storage site for DON [43], which can be transported with vertical water movement into shallow aquifers when irrigation and rainfall occur [42]. Furthermore, global reports indicate that DON concentrations in groundwater can range from 0.1 to 50.3 mg N L⁻¹, far exceeding the EU drinking water standards (≤ 1.0 mg N L⁻¹) [45]. The reactive transport and dispersion of DON in groundwater can contribute to regional nitrogen contamination [21], and its transformation into harmful by-products during the treatment of drinking water could increase cancer risks [2].

While progress has been made in investigating the distribution of DON [30], the geochemical processes governing its reactive transport in groundwater have only recently started to be explored. DON is a substantial nitrogen form in groundwater systems, playing a vital role in nitrogen cycling and enrichment [20]. Among various DON components, amino acids, urea and proteins are three key components that have been detected and identified in groundwater [28,34,11]. These components exhibit distinct structures and molecular weights, which in turn influence their reactive transport patterns. Amino acids, as the basic

building blocks of proteins, are relatively small molecules with diverse functional groups. Proteins are large, complex molecules with specific three-dimensional structures that can affect their mobility and reactivity in the environment. Urea is a small and highly soluble component. It is overused in agricultural activities, causing it to accumulate in large quantities in the vadose zone. Under frequent irrigation conditions, it has great potential to leach into aquifers [40]. As reactive solutes, these DON components interact with the aquifer microenvironment (including microorganisms and aquifer materials) as it migrates along groundwater flow paths, undergoing a series of biogeochemical transformations [45]. Various microorganisms with distinct nitrogen cycling functions can mineralize DON or use it as a carbon and energy source for other biochemical activities, leading to its continuous transformation into different nitrogen forms, which affects the nitrogen cycle and enrichment in groundwater [29]. For instance, urea has the potential to be mineralized in aquifers, leading to the formation of NO₃-N, which could exacerbate groundwater NO₃-N pollution. Amino acids and proteins could promote denitrification and temporarily alleviate NO₃-N pollution in groundwater [21]. In addition, abiotic processes such as adsorption and partitioning affect the distribution of DON between aqueous and solid phases [25], which in turn affects its mobility. Given their different structural and physicochemical properties, these DON components are expected to exhibit distinct reactive transport patterns in aquifers.

The current body of research has primarily focused on the vertical leaching process of DON through the vadose zone, particularly in agricultural cropping systems where intensive precipitation or irrigation can drive DON leaching from the soil to the deeper vadose zone [9]. For example, Lv et al. [24] examined the impacts of various irrigation practices on DON leaching. Wang et al. [42] analyzed the leaching behavior of DON and its potential mechanisms under different fertilization conditions. In addition, several studies utilized experimental data from soil columns to build the nitrogen transport model, simulating the vertical leaching behavior of DON in the vadose zone [28,10]. However, these studies have largely overlooked the broader reactive transport patterns of DON within aquifers under groundwater flow conditions. This gap in understanding limits our ability to fully comprehend the geochemical processes of DON throughout the entire groundwater system and to assess the environmental threats it may pose.

To address this knowledge gap, we investigated the reactive transport mechanisms of DON under groundwater flow conditions. We selected amino acids, urea and proteins to simulate DON components in groundwater, and conducted seepage tests using columns filled with aquifer materials, in order to provide insights into the reactive transport of different DON components in aquifers. Utilizing high throughput sequencing technology and bioinformatics analysis, we aimed to reveal DON's intrinsic role in the nitrogen cycle within aquifers. Owing to the distinct characteristics of DON, we hypothesized that distinct reactive transport patterns existed among different DON components. These variations were expected to exert a dual impact on NO₃-N enrichment in aquifers. The results of this study would enhance the understanding of geochemical processes involving DON and provide a more accurate assessment of the environmental risks associated with DON in groundwater.

2. Materials and methods

2.1. Chemicals

In this study, we selected urea, amino acids, and proteins as representative DON components (Table 1). These components have been previously detected and identified in groundwater samples [28,34,11]. Our selection is driven by their high bioavailability and mobility, making them ideal target components for experimental investigation. DON components were purchased from Sinopharm Co.

2.2. Study area and sampling

The aquifer materials and groundwater utilized in the experiments were obtained in Qingdao City, Shandong Province (N 36.697, E 120.238) (Fig.S1). The study area is characterized by a typical temperate continental monsoon climate, marked by hot and rainy summers as well as cold and dry winters. The annual average precipitation varies between 320 and 950 mm. The groundwater is primarily recharged by precipitation, river infiltration, and irrigation, and it discharges through runoff and agricultural use. The water table is located 1.5 m below the soil surface. Groundwater was extracted from a depth of 1 m below the water table by a stainless steel hydrophore (equipped with floating bottom plate and movable top cover), while the aquifer materials were obtained from a depth of 1.8–2.0 m below the surface using a soil drilling rig (Table 2). To maintain the integrity of the aquifer materials, they were transported to our laboratory under controlled temperature conditions and shielded from exposure to light. Following this, aquifer materials underwent a drying process and was sieved through a mesh with a 1 mm pore size to achieve the desired particle size for our experiment.

2.3. Column experiments

To simulate the reactive transport of DON, we used a 40 cm-long Plexiglas column with an inner diameter of 4 cm (Fig. 1). The experimental columns were positioned vertically and filled using the dry method. Stainless steel filter membranes with an inner diameter of 3 cm were set at both ends of the column to prevent particles from obstructing the pipeline. After adding each 5 cm-thick layer of aquifer materials, we compacted and compressed it using wooden sticks to avoid forming large pores or fissures in the column. To ensure uniform water distribution at the inlet and uniform effluent discharge at the outlet, a layer of quartz sand with a thickness of 2 cm and a particle size of 0.5 mm was placed at each end of the column. To minimize light interference, the entire experimental column was covered with aluminum foil. We then used a peristaltic pump to slowly saturate the column with ultrapure water from bottom to top, at a consistent flow rate. The water supply was interrupted for each 5 cm rise in head height in the column and maintained for a set period to eliminate air from the media's pores. We represented non-dimensional time as pore volumes, which is defined as the quotient of the column discharge and the water-filled pore-volume capacity of the packed column. According to the calculation, 1 PV $\approx$ 13.3 h in this study.

We conducted a tracer experiment to determine the flow regime in

Table 2

Selected physical and chemical properties of the water and aquifer samples.

	Indicators	Mean $\pm$ SD
Water sample	DO (mg L ⁻¹)	7.0 $\pm$ 0.21
	pH	7.4 $\pm$ 0.17
	Ec (ms cm ⁻¹)	2.4 $\pm$ 0.03
	NH ₄ ⁺ -N (mg L ⁻¹)	0.11 $\pm$ 0.02
	NO ₂ -N (mg L ⁻¹)	ND
	NO ₃ -N (mg L ⁻¹)	100.3 $\pm$ 0.81
	TDN (mg L ⁻¹)	101.0 $\pm$ 1.31
	DON (mg L ⁻¹)	0.6 $\pm$ 0.13
	DOC (mg L ⁻¹)	2.5 $\pm$ 0.57
	NH ₄ ⁺ -N (mg kg ⁻¹)	0.1 $\pm$ 0.01
Sediment sample	NO ₂ -N (mg kg ⁻¹)	ND
	NO ₃ -N (mg kg ⁻¹)	52.2 $\pm$ 0.08
	TDN (mg kg ⁻¹)	81.5 $\pm$ 0.13

Note: DO, dissolved oxygen. Ec, electrical conductivity. TDN, total dissolved nitrogen. DON, dissolved organic nitrogen. DOC, dissolved organic carbon. ND, not detected. Nitrogen concentrations in sediment samples were measured in 0.5 mol L⁻¹ K₂SO₄ solution (1:5 w/V).

the column and to provide a reference for the reactive transport of DON in the seepage column (Fig. S2). Sodium chloride was chosen as the tracer substance in the study. First, deionized water was continuously passed through the column for 48 hours to flush out salts and impurities on the surface of the aquifer materials. Subsequently, a 0.1 mol L⁻¹ sodium chloride solution (flow rate 0.2 mL min⁻¹) was introduced, and the electrical conductivity (Ec) of the effluent solution was regularly monitored. The tracer experiment was stopped when the electrical conductivity increased and reached a stable state. Finally, deionized water was continuously passed through to flush out the residual sodium chloride in the column.

To investigate the reactive transport patterns of three DON components in the aquifer, we prepared DON solutions (urea, amino acids, and proteins) using groundwater at a concentration of 30 mg L⁻¹. Specifically, we calculated the nitrogen concentration and chemical molecular weight of each DON component and weighed the corresponding weight of chemicals. After mixing the chemicals with the groundwater sample, we added it to a volumetric flask to prepare an experiment solution with an initial concentration of 30 mg L⁻¹. These prepared DON solutions, along with blank groundwater without added DON, were then passed through the column from bottom to top at a flow rate of 0.2 mL min⁻¹. Three replicates for each DON component reactive transport tests were conducted. We sampled the effluent from the top of all the columns every 1 PV and continuously monitored parameters such as DO (dissolved oxygen), ORP (oxidation-reduction potential), pH, NH₄⁺-N, NO₂-N, NO₃-N, and TDN (total dissolved nitrogen) concentrations. The test was concluded after introducing 20 PV (266 h) continuously. Moreover, we conducted transport tests for both non-sterilized and sterilized treatments to investigate transport patterns of DON under different conditions and quantitatively analyzed the effects of biotransformation and adsorption on DON transport. For the sterilized test, water bottles, aquifer materials and groundwater samples were subjected to high-temperature sterilization treatment in an autoclave at 121°C for 30 minutes, with the process repeated twice. Peristaltic pump hoses and columns are disinfected using ultraviolet radiation. This setup allowed us to study the abiotic processes and the intrinsic properties of DON without the influence of microorganisms. At the end of the experiment, we dismantled the columns and collected aquifer material samples from the upper, middle, and lower layers of the columns. The samples were then mixed to obtain a more representative sample of the entire soil column. Subsequently, DNA extraction and high-throughput sequencing were performed to identify the key functional microbial taxa involved in the reactive transport of DON in aquifers.

Table 1

Nature of DON components.

DON components	Molecular formula	Molecular weight	N content (%)	Purity
Urea	CH ₄ N ₂ O	60.1	46.6	Guaranteed reagent
Amino acids	C ₂ H ₅ NO ₂	75.1	18.7	Guaranteed reagent
Proteins	N/A	66.5 kDa	16.0	Guaranteed reagent

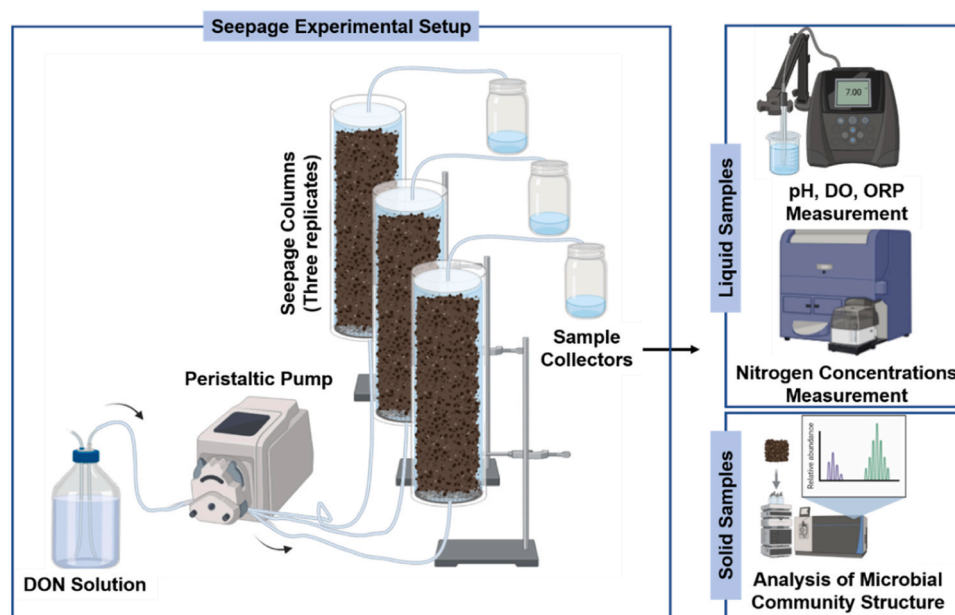


Fig. 1. Schematic diagram of the simulated seepage column tests.

2.4. Analysis of physicochemical properties

In this study, water samples were firstly passed through a 0.45 μm filter for the assessment of water quality indicators. DO, ORP and pH were determined by the Hach analyzer (HQ40d, Hach Inc., USA). $\text{NH}_4^+\text{-N}$ was analyzed using the Nesslerization colorimetric method. $\text{NO}_2^-\text{-N}$ was determined using the N-(1-naphthyl)-ethylenediamine photometric method. $\text{NO}_3^-\text{-N}$ was quantified using the UV spectrophotometric method. TDN was measured using the potassium persulfate oxidation-UV spectrophotometric method. DON was calculated using the difference subtraction method. All assessments of water quality indicators were performed in accordance with standard methods [8].

2.5. High throughput sequencing

DNA extraction from the solid phase was conducted using a MoBio PowerSoil® DNA Isolation Kit (MO BIO Laboratories, Inc., USA). The purity of DNA were assessed using a ultramicro spectrophotometer (Thermo Scientific, USA). Amplification of the V3-V4 region of the bacterial 16S rRNA gene was carried out using 338 F primers and 806 R primers [3]. Raw sequence analysis was conducted using the Illumina MiSeq paired-end 300 bp method (Illumina, USA). The resulting data were processed in QIIME 1.7.0, where quality filtered sequences were clustered into operable taxonomic units (OTUs) based on the 97 % sequence similarity of QIIME [5]. The analysis of bacterial community structure was further conducted using OTUs data.

2.6. Data analysis

Data visualization was performed using Origin 2017, and statistical analysis was conducted using SPSS Statistics 20. The results are presented as mean values $\pm$ standard errors ($n = 3$). Significant differences ($p < 0.05$) were analyzed by the LSD test. Online functional analysis of bacterial communities was performed using Majorbio.com. To enable functional annotation of biogeochemical cycling processes, we utilized the FAPROTAX database which links species classification to functional annotation. This database has been shown to have high prediction accuracy and is suitable for environmental samples with a focus on elemental cycles such as C, H, N, P, and S. It has previously been applied to environmental samples from various sources including oceans, lakes,

and soils [23].

3. Results and discussion

3.1. Dynamic changes of nitrogen concentrations during the reactive transport process

As shown in Fig. 2a, concentrations of $\text{NH}_4^+\text{-N}$, $\text{NO}_2^-\text{-N}$, and DON consistently remained at low levels ($< 0.5 \text{ mg L}^{-1}$), indicating negligible nitrogen release from the aquifer materials. $\text{NO}_3^-\text{-N}$ in the groundwater broke immediately through the column, increased until 4 PVs and thereafter remained constant at around 90 mg L^{-1} .

Following the flow of amino acids into the column, the DON concentration gradually increased and peaked at 9 PV, after which it fluctuated in the range of $20.7\text{--}24.1 \text{ mg L}^{-1}$ (Fig. 2b). The $\text{NH}_4^+\text{-N}$ concentration began to increase and peaked at 8 PV. Since there was no increase in $\text{NH}_4^+\text{-N}$ concentration in the blank treatment, the production of $\text{NH}_4^+\text{-N}$ in the amino acid column mainly originated from the mineralization of amino acids. The $\text{NO}_2^-\text{-N}$ concentration showed a slow increase during 7–20 PV. This phenomenon was attributed to the fact that the ammonia-oxidizing bacteria in the system had not yet formed dominant communities, resulting in the ammonia-oxidizing function not being fully established. In addition, due to its high mobility, the $\text{NO}_3^-\text{-N}$ concentration increased rapidly in the early period and peaked at 6 PV, followed by a decrease of 28.2 %. This may be due to the fact that amino acids contain a certain amount of organic carbon, and in the presence of amino acids, denitrification in the system was stimulated and intensified [48], leading to a decrease in $\text{NO}_3^-\text{-N}$ concentration.

In the urea transport experiment, the DON concentration in the groundwater steadily increased and peaked at 6 PV (Fig. 2c). The DON concentration was higher in the urea column because of its weak adsorption ability on porous media, which is confirmed by the previous research [22]. In the early stage of the transport test, the $\text{NH}_4^+\text{-N}$ concentration in the urea column was maintained at a low level. This was attributed to the lack of urease in the aquifer, which resulted in the inability to hydrolyze the urea. With time going on, the $\text{NH}_4^+\text{-N}$ concentration began to increase, indicating that urea began to undergo mineralization. This was the result of the continuous enrichment of urease-producing microorganisms [6]. In addition, the $\text{NO}_3^-\text{-N}$ concentration increased rapidly in the early stage of the experiment. At 5 PV, it

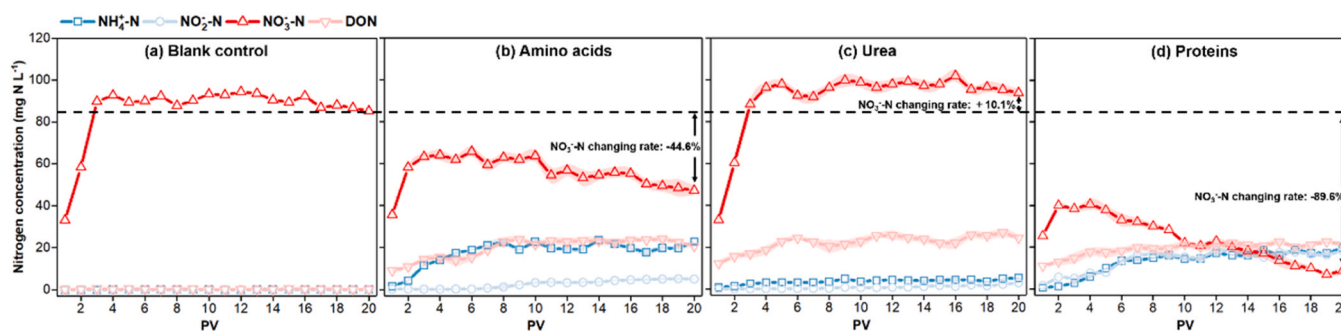


Fig. 2. Breakthrough curves of $\text{NH}_4^+\text{-N}$, $\text{NO}_2^-\text{-N}$, $\text{NO}_3^-\text{-N}$, and DON through columns filled with aquifer substrate percolated with (a) pure groundwater without amendment of DON, (b) groundwater plus 30 mg N L^{-1} glycine, (c) groundwater plus 30 mg N L^{-1} urea, (d) groundwater plus 30 mg N L^{-1} proteins. The results are presented as mean values $\pm$ standard errors ($n = 3$).

peaked and thereafter remained constant at around 100 mg L^{-1} . The $\text{NO}_3^-\text{-N}$ concentration of the urea column was 10.1 % higher than that of the blank treatment at the end of the experiment. This phenomenon can be attributed to the increasing microbial mineralization (ammonification-nitrification) capacity within the urea column. Over time, urea is gradually converted into $\text{NO}_3^-\text{-N}$ and continuously flow out of the column in inorganic form.

After the influx of proteins into the column, DON concentration gradually increased and peaked at 10 PV, subsequently fluctuating in the range of $20.2\text{--}22.7 \text{ mg L}^{-1}$ (Fig. 2d). The $\text{NH}_4^+\text{-N}$ concentration in the protein column exhibited rapid initial growth from 0 to 6 PV indicating mineralization of the protein, and then increased slowly. Additionally, $\text{NO}_2^-\text{-N}$ in the protein column showed a marked accumulation, with the peak $\text{NO}_2^-\text{-N}$ concentration occurring at 8 PV, reaching 16.8 mg L^{-1} . Along with the increase of the $\text{NO}_2^-\text{-N}$ concentrations, the $\text{NO}_3^-\text{-N}$ concentrations in the protein column showed a marked decrease. This was due to the fact that proteins also contain a large amount of organic carbon, which promotes denitrification in the column. The conversion from $\text{NO}_3^-\text{-N}$ to $\text{NO}_2^-\text{-N}$ is the first step of denitrification. $\text{NO}_3^-\text{-N}$ in the column was continuously converted to $\text{NO}_2^-\text{-N}$, which also explained the accumulation of $\text{NO}_2^-\text{-N}$ in the protein column. In addition, the ammonium-oxidation function of the protein column was not effectively established, resulting in the inability of $\text{NH}_4^+\text{-N}$ to be rapidly converted into $\text{NO}_2^-\text{-N}$. Therefore, it was inferred that the accumulation of $\text{NO}_2^-\text{-N}$ in the protein column mainly originated from the denitrification of $\text{NO}_3^-\text{-N}$. The decrease of $\text{NO}_3^-\text{-N}$ concentration in the protein column was greater than that in the amino acid column, which was attributed to the lower concentration of organic carbon in amino acids, resulting in a less denitrification in the amino acid column than in the protein column. This phenomenon is consistent with the prior finding that revealed the substantial denitrification potential of proteins [36]. Overall, under actual field conditions, variations in tillage practices and crop types result in significant differences in nitrogen (especially DON) loss from the vadose zone. For example, regions reliant on chemical fertilizers are at a higher risk of urea leaching, potentially leading to $\text{NO}_3^-\text{-N}$ pollution in regional groundwater. Conversely, areas predominantly utilizing organic fertilizers or straw may experience reduced $\text{NO}_3^-\text{-N}$ enrichment in groundwater due to the high organic carbon content in amino acids and proteins.

Moreover, we investigated variations in DO, ORP, and pH during the transport process of DON (Fig. 3). The DO concentrations decreased during the experiment in all treatments. This decline can be attributed to the oxidation reactions of organic matter, which consume dissolved oxygen in groundwater. Additionally, in all the DON treatments, ORP experienced a pattern of initial decrease, followed by an increase, and then another decrease. The fluctuations in ORP suggest a series of biogeochemical processes occurring within the columns. At the end of the transport experiments, the ORP values in three DON treatments were lower than the initial values, indicating a gradual trend towards a

reduction environment under the influence of microbial activity. Furthermore, the pH results revealed no marked fluctuations in pH values across the three DON treatments, due to the carbonate buffering system in groundwater.

3.2. Effect of biotransformation and adsorption on DON transport

As illustrated in Fig. 4, in the non-sterilized columns, the DON concentrations in the amino acid, urea, and protein treatments reached 23.9 , 25.8 , and 21.4 mg L^{-1} , representing 79.7 %, 86 %, and 71.4 % of the added DON concentrations. The effect extents of microbial transformation and adsorption to three DON components transport were 20.3 %, 14 % and 28.6 %, respectively. In the sterile columns, the DON concentrations in the amino acid, urea, and protein treatments stabilized at 26.3 , 28.6 , and 27.3 mg L^{-1} , respectively, which accounted for 87.6 %, 95.2 %, and 91 % of the added DON concentrations, respectively. The effect extents of adsorption to three DON components transport were 12.4 %, 4.8 %, and 9 %, respectively. It can be concluded that the effect extents of microbial transformation to three DON components transport were 7.9 %, 9.2 % and 19.6 %, respectively.

Combining the above results, it can be observed that during groundwater DON transport, adsorption exerts the greatest influence on the transport of amino acids, which is related to the strong adsorption capacity of the aquifer materials for amino acids. Besides, microbial transformation has the most important impact on protein transport, possibly because proteins can serve as an efficient denitrification carbon source and is thus extensively utilized by microorganisms [21].

3.3. Analysis of bacterial community characteristics

3.3.1. Analysis of species diversity indices

Ace and Chao indices in all DON treatments exceeded those observed in the blank control treatment (Fig. 5). This suggests that the transport of the three DON components (amino acids, urea, and proteins) in groundwater can enhance the abundance of bacteria, because DON can serve as a nutrient source and promote microbial growth. Furthermore, the Simpson and Shannon indices of all DON treatments were higher than those of the blank control treatment, indicating an increase in microbial diversities. Generally, increasing nutrient concentrations can reduce the biodiversity. However, the observed increase in microbial diversity following DON addition may be attributed to the fact that DON stimulated the activity of different functional species, particularly nitrogen cycling microorganisms, further enhancing the microbial diversity, aligning with the previous research [37]. In addition, the aquifer materials were initially nutrient-limited, and DON addition alleviated this limitation, allowing previously suppressed microorganisms to grow and multiply. This is consistent with the previous finding, which have shown that moderate nutrient addition in nutrient-limited environments can alleviate competition among microbial communities, leading to the

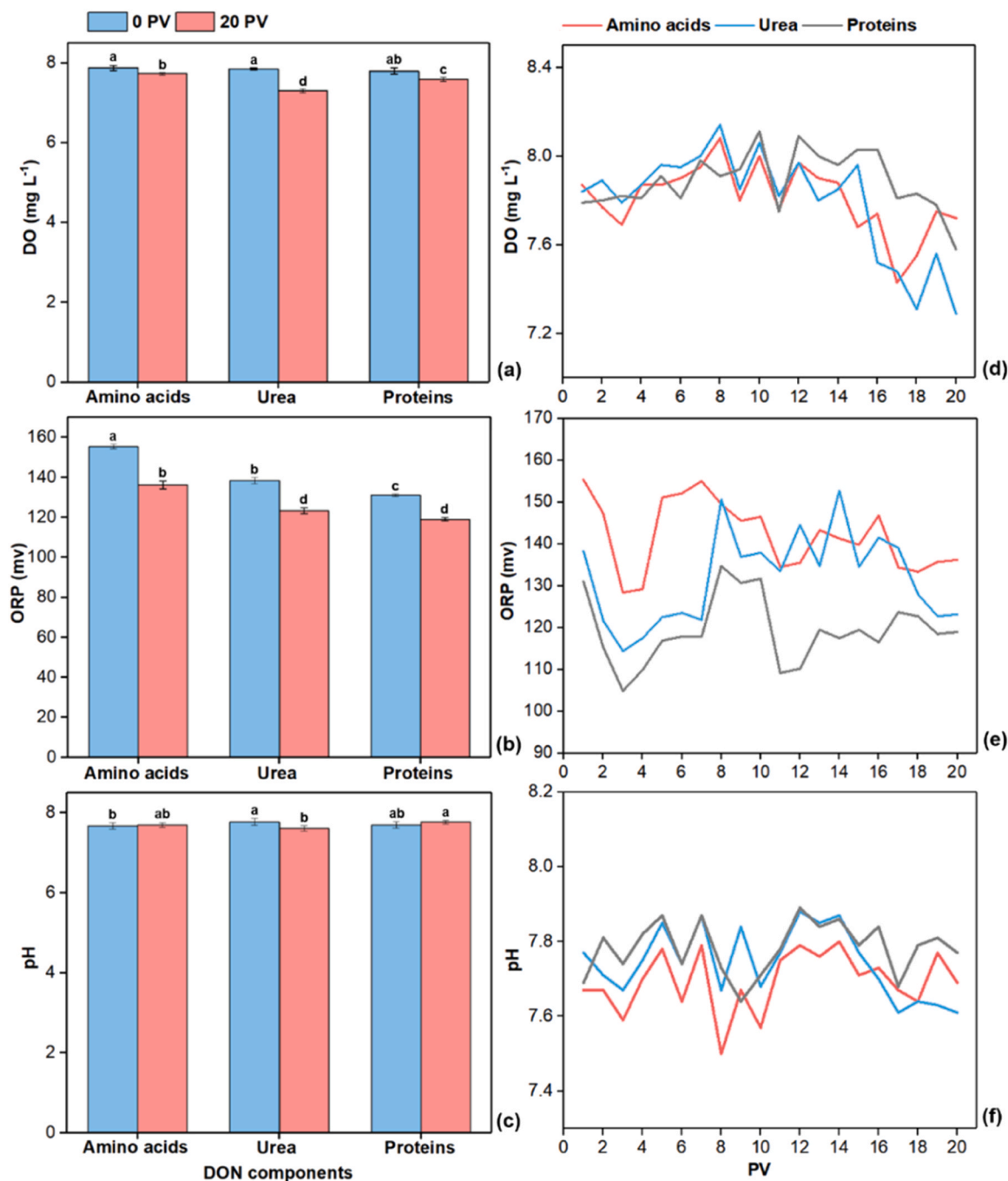


Fig. 3. Values of DO, ORP and pH at 0 and 20 PV of three DON treatments (a-c). DO, ORP and pH dynamic curves of DON columns (d-f). The results are presented as mean values $\pm$ standard errors ($n = 3$). Different lowercase letters indicate significant differences ($p < 0.05$) as determined by the LSD test.

increased diversity [41].

3.3.2. Analysis of microbial community structure

At the phylum level, the transport of DON in aquifers influences the microbial community structure, leading to increased relative abundances of Proteobacteria, Actinobacteria, Acidobacteria, and Chloroflexi (Fig. 6a). Among these phyla, Proteobacteria emerged as the most dominant group, exhibiting the most pronounced increase in relative abundance. This phenomenon can be attributed to the involvement of Proteobacteria in nitrogen cycling. Proteobacteria encompass a diverse array of microorganisms, including nitrogen-fixing bacteria, ammonia-oxidizing bacteria, cellulose-degrading bacteria, nitrifying bacteria, and denitrifying bacteria. Each of these microbial groups plays a pivotal

role in nitrogen transformation processes, thereby facilitating the cycling of nitrogen in the environment [52]. In addition to the observed increase in Proteobacteria, Actinobacteria also exhibited enrichments in all DON treatments. Actinobacteria are well-known for their strong ability to decompose organic matter [46], which plays a critical role in the ammonification process (organic nitrogen $\rightarrow$ $\text{NH}_4^+\text{-N}$). The presence of DON in groundwater tends to enhance the relative abundance of Actinobacteria, thereby potentially stimulating the ammonification process, ultimately contributing to the production of $\text{NH}_4^+\text{-N}$ in groundwater. This finding is consistent with previous studies that have highlighted the role of Actinobacteria in nitrogen cycling processes [4].

At the genus level (Fig. 6b), relative abundances of *Pseudomonas*, *Blastocatellaceae*, and *Thermomonas* were found to increase in all DON

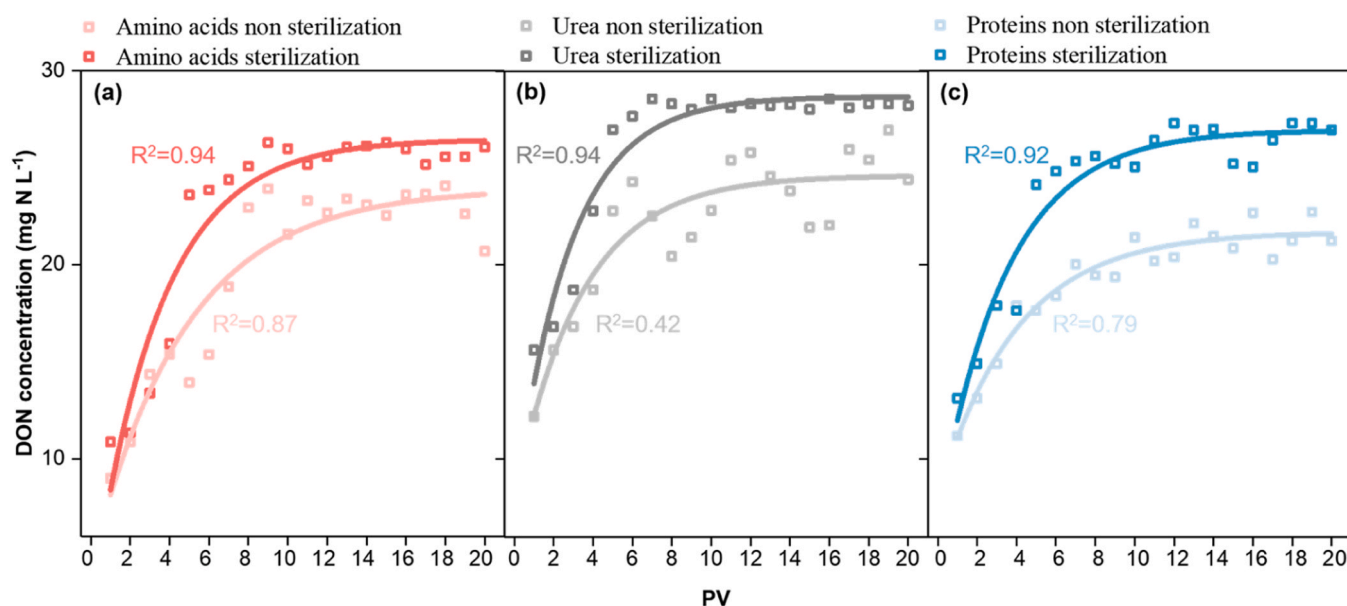


Fig. 4. Fitting results of DON breakthrough curves through columns filled with aquifer substrate percolated with (a) groundwater plus 30 mg N L⁻¹ glycine, (b) groundwater plus 30 mg N L⁻¹ urea, (c) groundwater plus 30 mg N L⁻¹ proteins under non-sterilized and sterilized conditions.

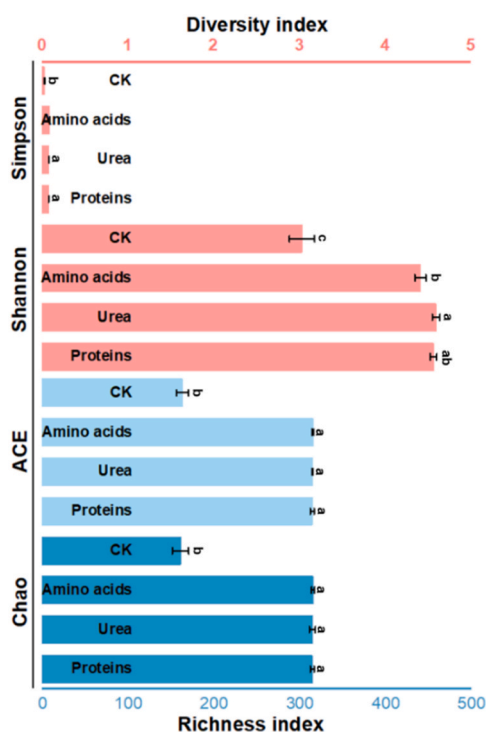


Fig. 5. Species richness indices and diversity indices of DON treatments. The results are presented as mean values $\pm$ standard errors ($n = 3$). Different lowercase letters indicate significant differences ($p < 0.05$) as determined by the LSD test.

treatments. *Pseudomonas*, emerging as the most dominant genus, exhibited higher relative abundances in amino acid (10.1 %) and protein (7.3 %) treatments, compared to the urea treatment. *Pseudomonas*, a typical Gram-negative bacterium that is widely distributed subsurface sediments, has been previously recognized for its important role in the transformation of organic matter [44]. Furthermore, certain strains of *Pseudomonas* possess aerobic denitrification capabilities [15,18]. These functional characteristics of *Pseudomonas* suggest that its increased

relative abundance in amino acid and protein columns may enhance the efficiency of denitrification in groundwater. Moreover, the relative abundance of another bacterial genera with denitrifying function, *Thermomonas*, was also higher in amino acid (3.9 %) and protein (5.1 %) treatments. A previous research has shown that the addition of organic matter can promote the biological activity of *Thermomonas*, with NO₃-N serving as electron acceptor for denitrifiers, thereby causing the denitrification [39]. Therefore, the introduction of amino acids and proteins into aquifers as sources of organic carbon has been shown to promote in situ denitrification, thereby reducing the risk of NO₃-N pollution. This molecular biological analysis validates the previous finding that the NO₃-N concentration is lower in the presence of amino acids and proteins columns.

In addition, in the urea column, the relative abundances of bacteria with nitrification function were higher than those in amino acid and protein columns. Specifically, the relative abundance of *Nitrosomonadaceae*, which has ammonia oxidation function, was 0.18 %–0.19 % higher in the urea column. And the relative abundance of *Nitrospira*, which has nitrite-oxidation function, was 0.07 %–0.15 % higher in the urea column. Despite the relatively low abundances of *Nitrosomonadaceae* and *Nitrospira* within the overall microbial community, these genera are essential for the nitrification process in natural environment [13]. Our analysis reveals that the presence of urea in groundwater promotes the activity of *Nitrosomonadaceae* and *Nitrospira*, thereby enhancing the nitrification process. This enhancement is consistent with the observed increase in NO₃-N concentration in the urea column, supporting the conclusion that urea transport in aquifers can exacerbate the risk of NO₃-N pollution in regional groundwater.

3.3.3. Microbial function prediction

Employing the FAPROTAX functional prediction module, we examined variations in microbial community functions across diverse DON treatments (Fig. 7). The FAPROTAX prediction results showed an increase in the average abundance of (aerobic) chemoheterotrophic bacteria in three DON treatments compared to the blank treatment. This suggests that the presence of DON supplies substrates for microbial activities in aquifers.

Furthermore, the result demonstrated that microorganisms involved in nitrogen cycling, particularly those mediating denitrification, were predominant in DON columns. This dominance is likely attributable to

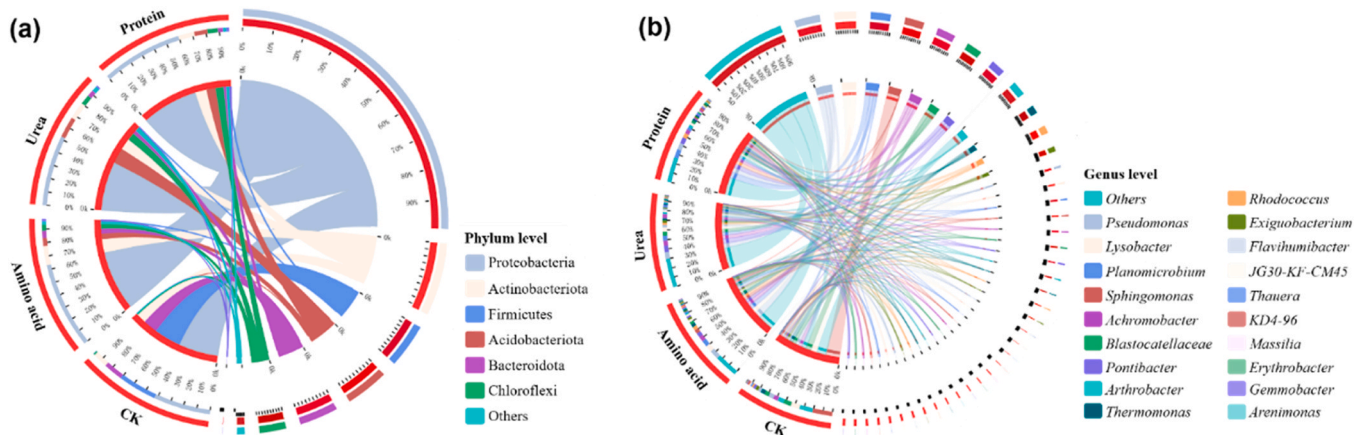


Fig. 6. Bacterial community characteristics of DON columns. (a) Circos representation of dominant bacterial communities at phylum level. (b) Circos representation of dominant bacterial communities at genus level.

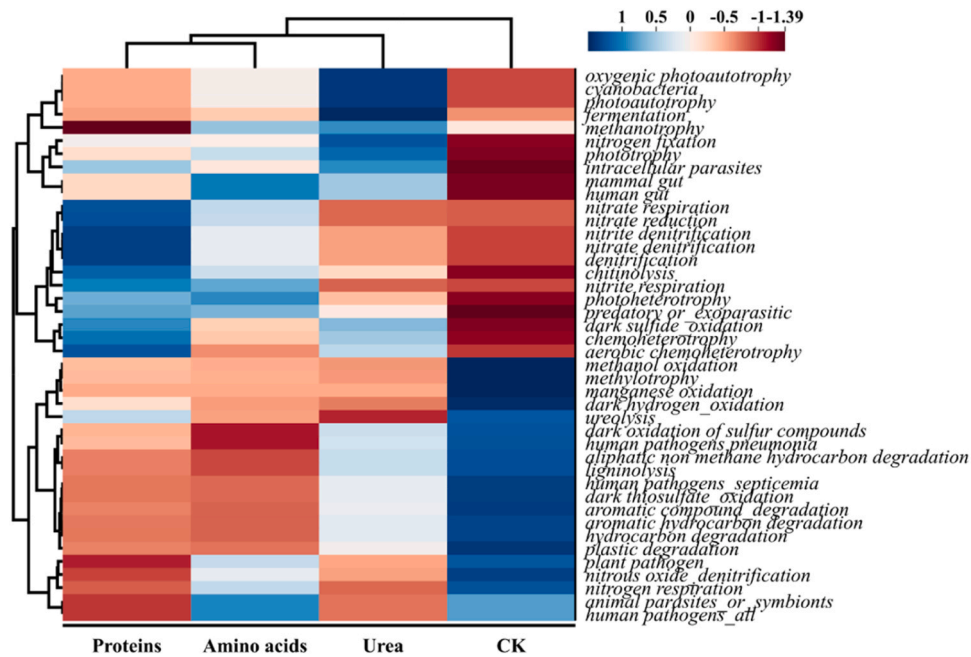


Fig. 7. Heat map of functional bacteria in different DON columns. The color ranging from red to blue indicates an increase in relative abundances of the functional bacteria.

the addition of DON, which provides essential carbon and energy sources for heterotrophic denitrifying microorganisms. It was found that the relative abundances of microorganisms involved in denitrification was higher in the protein column compared to the amino acid and urea columns. This observation suggests that proteins provide a more substantial supply of organic carbon sources to support denitrification processes, which is consistent with the previously observed phenomenon of a lower $\text{NO}_3\text{-N}$ concentration in the protein column. However, nitrite-reducing and nitrogen oxide-reducing microorganisms were less abundant than nitrate-reducing microorganisms. This observation may be explained by incomplete denitrification process resulting from insufficient organic carbon content in DON. Consequently, the reduction of $\text{NO}_2\text{-N}$ and N_2O could become the rate-limiting steps for $\text{NO}_3\text{-N}$ denitrification during the reactive transport of DON in aquifers [7].

4. Conclusion

We explored the reactive transport and fate of DON in aquifers to

reveal the differences in reactive transport patterns among various DON components and their impacts on $\text{NO}_3\text{-N}$ accumulation in groundwater. The results show that urea exhibits greater mobility, indicating a higher nitrogen contamination risk. During the reactive transport process, urea leads to the accumulation of $\text{NO}_3\text{-N}$ in groundwater, while amino acids and proteins cause a reduction of $\text{NO}_3\text{-N}$ concentration. Therefore, it is essential to pay particular attention to the nitrogen pollution status of groundwater in areas where urea leaching occurs. This study also provides information at the molecular biology level, that is, the reactive transport of DON can increase the relative abundance of bacteria with specific nitrogen cycling functions (*Pseudomonas*, *Thermomonas* and *Nitrospira*), which can explain the changes in $\text{NO}_3\text{-N}$ concentration during different DON transport processes. In addition, biotransformation has the greatest effect on protein transport, while adsorption mainly influences amino acid transport.

Environmental implication

Dissolved organic nitrogen is often an overlooked form of nitrogen that can leach from the soil into aquifers. The reactive transport and dispersion of DON in aquifers can contribute to regional nitrogen contamination. The study investigates the reactive transport mechanisms of DON under groundwater flow conditions. We discover that the transport of amino acids and proteins causes a reduction of $\text{NO}_3\text{-N}$ concentration, while urea leads to the accumulation of $\text{NO}_3\text{-N}$ in groundwater. Urea exhibits greater mobility, indicating a higher nitrogen contamination risk. These findings hold important implications for regional groundwater environment protection.

CRediT authorship contribution statement

Liu Yang: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zheng Tianyuan:** Writing – review & editing, Supervision, Resources, Methodology, Investigation, Conceptualization, Funding acquisition. **Guo Bo:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Tao Yiheng:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Jiang Shiqiang:** Supervision, Resources, Investigation. **Cao Min:** Supervision, Resources. **Zheng Xilai:** Supervision, Conceptualization. **Luo Jian:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jhazmat.2025.138259](https://doi.org/10.1016/j.jhazmat.2025.138259).

Data availability

Data will be made available on request.

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